Digital Library

Introduction

A digital library was developed as a result of the rapid growth of information and advances in Information and Communication Technology (ICT). Traditional libraries, which mainly rely on printed materials, are no longer sufficient to meet the increasing demand for quick and remote access to information. Digital libraries therefore, emerged to complement conventional libraries by providing electronic access to information resources. They support teaching, learning, research, and information sharing in the modern knowledge-driven society.

Meaning of Digital Library

A digital library is a library system in which information resources are stored, organized, and accessed in digital or electronic format using computers and network technologies. These resources include e-books, e-journals, online databases, digitized theses and dissertations, images, audio, and video materials. Digital libraries allow users to access information without physical contact with the library building, making information available anytime and anywhere through the internet.

Components of a Digital Library

Digital Collections

Digital collections are the core contents of a digital library. They consist of information resources available in electronic format, including born-digital materials such as online journals and databases, as well as digitized materials converted from print to digital form. These collections are selected based on users' needs and institutional goals, ensuring relevance and usefulness for learning and research.

Technological Infrastructure

Technological infrastructure refers to the hardware, software, and network facilities that support digital library operations. Hardware includes computers, servers, scanners, and storage devices,

while software includes digital library systems and repository platforms. Network facilities such as the internet enable remote access to resources. Without adequate infrastructure, digital libraries cannot function effectively.

Human Resources

Human resources are the trained professionals responsible for managing digital libraries. They include librarians, information scientists, and IT specialists who handle resource selection, organization, system maintenance, and user support. Their skills ensure that digital resources are properly managed, accessible, and preserved for long-term use.

Policies and Standards

Policies and standards guide the operation of digital libraries. Policies address issues such as access rights, copyright, and ethical use of information. Standards such as metadata schemas ensure consistency in resource description and enable resource sharing across different digital platforms. These policies and standards promote legality, organization, and sustainability.

Open Access

Introduction

Open Access (OA) emerged as a response to the rising cost of scholarly publications and the restrictions placed on access to academic research. Traditionally, scholarly articles and research outputs were locked behind subscription fees, limiting access to only those who could afford them. Open Access was introduced to remove these barriers and ensure that research findings are freely available to anyone with internet access. It plays a vital role in promoting knowledge sharing, academic visibility, and global research development.

Meaning of Open Access

Open Access refers to the free, immediate, and unrestricted online availability of scholarly research outputs such as journal articles, conference papers, theses, and research reports. Under Open Access, users can read, download, copy, distribute, and use research materials without financial,

legal, or technical barriers, provided proper acknowledgment is given to the author. Open Access aims to make knowledge a public good rather than a commodity restricted to paying subscribers.

Types of Open Access

Gold Open Access

Gold Open Access refers to scholarly articles that are made freely available by the publisher immediately upon publication. These articles are published in Open Access journals, and users can access them without subscription fees. In some cases, authors or their institutions may be required to pay Article Processing Charges (APCs) to cover publication costs. Gold Open Access ensures immediate and permanent access to research outputs.

Green Open Access

Green Open Access involves the self-archiving of research outputs by authors in institutional repositories or subject repositories. In this model, authors publish their articles in traditional subscription-based journals but also deposit a copy of the article in an open repository. Access may be immediate or delayed due to embargo periods imposed by publishers. Green Open Access increases access while allowing authors to publish in conventional journals.

Hybrid Open Access

Hybrid Open Access is a model in which subscription-based journals offer authors the option to make individual articles openly accessible upon payment of an APC. While the journal itself remains subscription-based, selected articles are freely available to users. This model combines features of both traditional and Open Access publishing systems.